

**Delta RMSE from adding disturbance – ten variants: RF/XGBoost/LightGBM, each untuned, grid-tuned and Optuna-tuned (+ RF exp01/exp02)**

Mean % change in LOSO RMSE when the disturbance block D is added, averaged over the 5 EFPs.  
NEGATIVE = less error = disturbance helps. M5–M8 split by memory type. Same data, same variables, same folds.

